Ram Calculus Audit of the Quasi-Riemann Preprint

Exact Endpoint Optimization, Capacity Bounds,
and Conditional Mellin Interfaces

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Abstract

This report audits selected mathematical interfaces and the full finite endpoint calculation in the pinned OpenAI preprint claiming a zero-free half-plane $\Re s > 7/8$ for finite-order Hecke functions over $\mathbb{Q}(\sqrt{-3})$ and for Dirichlet functions. It does not certify that global claim. The classical Mellin continuation mechanism is rederived with its exact uniformity requirements, and the theta, prime, moment, and normalization inputs are separated from the conclusions they would imply. Using exact tensor Bernstein certificates, we prove that the source's two-variable endpoint margin has unique minimum $(-4473 + 35\sqrt{16354})/12696$, strictly between $1/4384$ and $1/4383$. The minimum was identified in an earlier local audit; here it receives a standalone proof, independent coefficient checks, and explicit stability and perturbation bounds. Retaining the source's dynamic bin constraint also improves its finite capacity penalty from $\Delta/4$ to $4\Delta/27$, conditionally changing the endpoint comparison coefficient from $51/64$ to $95/108$. These are bounded improvements to one proof component, not a new zero-free region or a proof of the Riemann hypothesis. The report gives a local-corpus provenance map, a current primary-literature check, exact reproducibility evidence, limitations, and a prioritized research program. Contribution assessment: C2/5, exact bounded improvements and auditable interfaces without a new global analytic theorem.

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1 Scope, source identity, and claim discipline

The object of this report is OpenAI's *The Quasi-Riemann Hypothesis: A Zero-Free Half-Plane* $\Re(s) > 7/8$, dated 30 September 2026 [1]. The authoritative input here is its actual `build/paper.tex`, not the release overview or a title-level catalogue entry. The repository pin is

`adc7f1241b42e322a6451854ab7e4b4c146bf78a`.

The local source SHA-256 is

`42a5ee0febca59fd1def55cfd6c6808c322e`
`f4237d7726303ea52f711deac6a3`.

A read-only HTTP retrieval of the public raw file at that commit returned status 200 and 766316 bytes, with exactly the same hash and bytes. The local source directory is an exported snapshot, not a Git checkout; a failed `gitrev-parseHEAD` is not treated as pin verification. The byte comparison is the verification. All private corpus passages stayed local.

The source identifies itself as an *OpenAI Math Release preprint*. Neither its abstract, a repository catalogue, nor a listed formalization is presumed to establish acceptance or correctness. This author report supplies a complete proof of its own bounded results and an explicit interface audit, not an independent certification of the entire analytic number-theory chain. The completed independent analytic review found no established blocking defect in its checked portions, but did not independently verify every reflection, moment recursion, or uniformity assertion. Its bounded outcome is integrated in Appendix A.1; it is not a proof endorsement.

Status	Meaning in this report
Source claim	The preprint asserts the 11/12 and then 7/8 half-plane theorems. We describe them as assertions under review.
Imported input	Classical Fourier/Mellin inversion, finite-field character orthogonality, Hecke theory, and fixed-extension prime counting are used only with the hypotheses specified below.
Reproved interface	The common-signal implication, local Fourier-mask identities, and elementary height/loss closure are proved explicitly here; their arithmetic premises are separate obligations.
Proven local result	The finite rational minimum, minimizer localization, perturbation tolerance, and capacity comparison are mathematical results of this report, independently of the truth of the source's global theorem.
Conditional bridge	Replacing an endpoint constant in the source's high estimate is valid only if its detector, moments, masks, contour moves, normalizer, and uniformity assertions all hold.
Not established	No new L-function zero-free half-plane, RH, GRH, exceptional-zero exclusion, or global validity certificate.

Applicability L4 describes how directly Ram transform and polynomial certificate machinery engages this source. It is not a validity verdict, and it is not the contribution score C2. The most substantive new increment beyond the earlier finite audit is the dynamic capacity estimate in Section 10, with the quantitative bounded extensions in Section 9. The exact minimum itself is credited to the earlier local audit, whose code and proofs were read and checked.

2 The source's architecture and its exact obligations

Put $F = \mathbb{Q}(\sqrt{-3})$ and let $\mathcal{O}$ be its integers. For a primitive finite-order Hecke character η, extended by zero at its ramified ideals, the Euler product defines $L_F(s,\eta)$ for $\Re s > 1$ and classical Hecke theory supplies meromorphic continuation. Define

$$\beta_* = \sup\big(\{1/2\} \cup \{\Re\rho : L_F(\rho,\eta) = 0,\ 1/2 \le \Re\rho \le 1,\ \eta \text{ primitive finite-order}\}\big). \tag{2.1}$$

This is a supremum over the *whole family*, including all heights and conductors. It need not be attained. Poles are not included.

The first source stage uses balanced scales and the Mellin power $C_{\mathrm{I}}(s) = s - 2/3$, aiming at $\beta_* \le 11/12$. Its second stage assumes that first conclusion and, for a contradiction, sets

$$0 < \Delta = \beta_* - 7/8 \le 1/24, \qquad \kappa = 2\beta_* - 1 = 3/4 + 2\Delta \le 5/6. \tag{2.2}$$

The compensated geometry is

$$h = \frac{13}{16}, \quad \ell = \frac{1}{6}, \quad l_x = \frac{17}{48}, \quad l_y = \frac{23}{48}, \quad h = 1 - l_x + \ell, \quad C_{\mathrm{II}}(s) = s - \frac{11}{16}. \tag{2.3}$$

In particular $C_{\mathrm{II}}(7/8) = 3/16$ and $C_{\mathrm{II}}(\beta_*) - C_{\mathrm{II}}(7/8) = \Delta$. The two stages have different physical normalized sums. They are not interchangeable just because they use the same continuation proposition.

2.1 Dependency map from the actual manuscript

The following source labels are stable locators in the pinned TeX. Line ranges for the author reads are recorded in the evidence ledger.

Source component	Required mathematical content
`prop:completed-reflection`	The completed indices are nb^3, n squarefree; no condition $(n,b) = 1$ is inserted. Whole-index characters retain their zero values, even at exponent zero. Rational cusp phases and the three fixed coefficient families must match simultaneously.
`lem:local-euler`	The Poisson row quotient must have the stated denominator and a holomorphic correction; the principal residue must contain the target reciprocal with a nonvanishing multiplier.
`prop:detector-witness`	The inverse and plain witnesses have one common row character and one common twist height. Bounded-height buffers, the floor class, and all admissible presentations must be retained.
`lem:marked`	With row width m, inverse length r, and total prime length z, the strict bounds are $r + 2z \le m - c_1$ and $2r + 8z \le 3m - c_2$ for fixed $c_1, c_2 > 0$. Prime sets and coefficients are row-independent.
`lem:plain`	Positive slots require $n_1 + n_2 + 6\kappa z \le M$ and, if $\kappa < 1$, $\beta_* \le (1+\kappa)/2$. The slot mesh is pretarget; exceptional inducing characters in Θ are excluded differently from zero-slot principal rows.

`prop:detector-counts`	The row-count envelope requires a supply strictly exceeding 7/37, slot selection margins, and height factors; its scalar optimization is proved below, not its analytic input moments.
`lem:high-bin`	Move the full holomorphic fixed-bin correction before pointwise amplitude subdivisions. Use a supremum over occurring real exponents; do not integrate nonholomorphic pointwise pieces separately.
`prop:parameter-order`	Positive final exponent margins precede the target. Fixed arithmetic data, internal orders, height exponent, external orders, and sufficiently large thresholds follow in that order.

The source's inverse and plain recursions are substantial arithmetic theorems, not applications of Bernstein positivity. Our evidence ledger records a full read of the completed-reflection proposition and its proof, the moment statements, the prime-estimate passage, and the row-count and conclusion calculations. It does not claim an independent reproduction of all recursive Poisson estimates. A source-specific audit can distinguish a correctly proved conditional interface from an unverified premise.

2.2 Transfer to Dirichlet functions

Conditional on the Hecke half-plane, the final transfer is ordinary quadratic base change. Away from a fixed finite set of Euler factors,

$$L_F(s, \chi \circ N) = L(s, \chi)L(s, \chi\chi_{-3}).$$

At a split prime the two degree-one local factors coincide; at an inert prime the product $(1 - ap^{-s})^{-1}(1 + ap^{-s})^{-1}$ equals $(1 - a^2p^{-2s})^{-1}$. Ramified and conductor primes are handled by finite factors. For $\Re s > 0$ such factors are nonzero, since $|\chi(p)p^{-s}| < 1$ whenever the local character value is nonzero. Away from $s = 1$ there is no pole of the companion factor to hide a zero. At $s = 1$, the usual nonvanishing theorem for nonprincipal Dirichlet functions and the principal simple pole handle the exceptional point. This is a transfer *from* a Hecke theorem; it supplies no proof of the Hecke theorem itself.

3 Actual private-corpus interfaces

Four local passages are used, for different and limited purposes. No corpus title is treated as evidence for a theorem.

Volume 2, Chapter 1. The section `v2:c01:mellin`, especially `v2:c01:mellin-log` and the theorem "Mellin inversion as weighted Fourier inversion," derives the transform by $x = e^u$ and states the $L^1 \cap L^2$ hypotheses. The report's Ram-Master interpretation uses this exact logarithmic coordinate change. It does not attach an automatic inverse or a zero-free consequence to an arbitrary atom bank.

Volume 16, Chapter 5d. The power-to-Bernstein lemma `v16c05:conversionproof`, convex-hull proposition `v16c05:hull`, and exact subdivision passage supply a finite certificate method. They explicitly distinguish a positive coefficient certificate from point sampling, and an inconclusive negative coefficient from a negative polynomial. Section 8 extends the displayed identity to two variables by tensor products.

Revised TR250. "What a Ram representation means in this report," including `eq:moments`, writes $m = Ec$ on a specified finite space and requires injectivity and a left inverse. "A concrete polynomial Ram construction" proves unisolvence from a positive weighted Gram matrix. We use precisely that finite-space discipline: the endpoint numerator is in $\mathbb{P}_{2,3}$, and its derivative certificate is in $\mathbb{P}_{2,4}$. An invertible coordinate change neither proves positivity nor improves the underlying function by itself.

TR220, version 3. The actual local-statistics barrier passage (lines 304–317) warns that finite or density-based information cannot exclude every exceptional zero. Its explicit-formula passage (lines 534–559) places the universal quantifier on *every admissible test*, with controlled limits. We use that warning, not its separate critical-line-proportion programme, as the limit on what a finite endpoint certificate can establish. No private RH theorem is imported into the present proof.

For example, on a polynomial rectangle B with basis ϕ_j, moments $m_i = \int_B w\phi_i p$ give a positive definite Gram matrix when $w > 0$ almost everywhere. Indeed $c^*Ec = \int_B w|\sum c_j\phi_j|^2 > 0$ for nonzero c. This reconstructs p exactly from compatible moments. The subsequent conversion to Bernstein coefficients is another exact finite map. The *signs* of those coefficients, not the word "Ram," give the enclosure. The report's executable checks use rational grid interpolation as an independent coordinate realization, not a numerical fit to unknown infinite-dimensional data.

4 Mellin atoms and a rigorous continuation interface

4.1 Measures, signs, and the Gaussian

For a suitable function V on $(0,\infty)$,

$$\widehat{V}(s) = \int_0^\infty V(x)x^s\,\frac{dx}{x} = \int_{\mathbb{R}} V(e^u)e^{su}\,du. \tag{4.1}$$

Thus, with Fourier convention $\mathcal{F}g(t) = \int g(u)e^{-\mathrm{i}tu}\,du$, $\widehat{V}(c+\mathrm{i}t) = \mathcal{F}(e^{cu}V(e^u))(-t)$. The sign and the measure dx/x matter. An extra Jacobian would shift the Mellin argument by one. Under weighted $L^1 \cap L^2$ hypotheses, Fourier inversion and Plancherel give

$$V(x) = \frac{x^{-c}}{2\pi}\int_{\mathbb{R}} \widehat{V}(c+\mathrm{i}t)x^{-\mathrm{i}t}\,dt, \qquad \int_0^\infty |V(x)|^2x^{2c-1}\,dx = \frac{1}{2\pi}\int_{\mathbb{R}} |\widehat{V}(c+\mathrm{i}t)|^2\,dt.$$

The inverse is weighted-L^2 unless stronger pointwise assumptions hold. These are classical transform theorems [2], reexpressed in the local corpus's atom notation, not new inversion theorems.

The source uses

$$V(x) = \frac{1}{2\sqrt{\pi}}\exp\bigl(-(\log x)^2/4\bigr), \qquad \widehat{V}(s) = e^{s^2}.$$

Substitution in (4.1) and completion of the square proves the identity first for real s; exponential domination on compact subsets then extends it to complex s. Its logarithmic derivatives have rapid power decay at both endpoints. On a vertical line, $|e^{(c+\mathrm{i}t-5/6)^2}| = e^{(c-5/6)^2-t^2}$, supplying height decay.

Proposition 4.1 (Common-signal continuation, with its quantifiers)**.** *Assume the classical analytic theory of the above L-functions. Fix $1/2 < \sigma_0 < \beta_*$ and $C(s) = s + c$. Suppose there are $0 < \omega < \beta_* - \sigma_0$ and $\sigma > 0$, independent of η, such that for each primitive target there exist a finite set S, a holomorphic H_η on $\Re s > \sigma_0$ with*

$$\sup_{\Re s > \sigma_0} |H_\eta(s) - 1| \le 1/2, \tag{4.2}$$

and a function J_η defined for all sufficiently large real Z, for which

$$f_\eta(Z) = \frac{1}{2\pi \mathrm{i}} \int_{(2)} Z^{s+c} e^{(s-5/6)^2} \frac{H_\eta(s)}{L_F^S(s,\eta)}\, ds, \tag{4.3}$$

$$|J_\eta(Z)| \ll_\eta Z^{\sigma_0+c+\omega}, \qquad |J_\eta(Z) - f_\eta(Z)| \ll_\eta Z^{\beta_*+c-\sigma}. \tag{4.4}$$

Here constants and lower thresholds may depend on the target. These assumptions contradict $\beta_* > \sigma_0$.

Proof. Set $\epsilon_* = \min(\beta_* - \sigma_0 - \omega, \sigma) > 0$. The triangle inequality yields $f_\eta(Z) = O_\eta(Z^{\beta_*+c-\epsilon_*})$ as $Z \to \infty$. Since the reciprocal Euler product is bounded on $\Re s \geq 2$ and H_η is bounded there, the Gaussian permits a rightward shift in (4.3) to any fixed $B > 2$. Horizontal integrals vanish by Gaussian decay on the finite strip. Consequently $f_\eta(Z) = O_{\eta,B}(Z^{B+c})$ for $0 < Z \leq 1$.

For $\Re s > \beta_* - \epsilon_*$ the integral

$$F_\eta(s) = \int_0^\infty f_\eta(Z) Z^{-s-c} \frac{dZ}{Z}$$

converges locally uniformly. Near infinity the exponent is less than zero by a compact-set margin; near zero choose B larger than all real parts on the compact set. Logarithmic factors arising in derivatives are absorbed by slightly smaller margins, so F_η is holomorphic.

Let $A_\eta(s) = e^{(s-5/6)^2} H_\eta(s)/L_F^S(s,\eta)$. On the line $2 + \mathrm{i}t$ it is continuous and integrable. Writing $Z = e^u$ gives

$$e^{-(2+c)u} f_\eta(e^u) = \frac{1}{2\pi} \int_{\mathbb{R}} e^{\mathrm{i}tu} A_\eta(2 + \mathrm{i}t)\, dt.$$

The left side belongs to $L^1(\mathbb{R})$ by the two endpoint bounds just proved. Fourier inversion therefore identifies its transform with the continuous function $A_\eta(2+\mathrm{i}t)$. Thus $F_\eta = A_\eta$ on that line, and the identity theorem extends equality to $\Re s > 1$.

By (4.2), $|H_\eta| \geq 1/2$, so $e^{-(s-5/6)^2} F_\eta(s)/H_\eta(s)$ continues the reciprocal holomorphically to $\Re s > \beta_* - \epsilon_*$. This half-plane lies inside $\Re s > \sigma_0$. A finite deleted factor $1 - \eta(\mathfrak{p}) N\mathfrak{p}^{-s}$ has no zero there. By the supremum definition, some target has a zero with real part greater than $\beta_* - \epsilon_*$; the reciprocal has a pole there, contradicting the continuation. No maximum-zero assumption was used. At a principal pole $s = 1$ the reciprocal has a zero, which causes no obstruction. □

The required quantifier order is

$$\exists \omega, \sigma > 0 \quad \forall \eta \quad \exists S_\eta, H_\eta, J_\eta, Z_{0,\eta}, K_\eta \quad \forall Z \geq Z_{0,\eta}.$$

The weaker order $\forall \eta\, \exists \sigma_\eta > 0$ does not give a common gap below a family supremum. Bounds on a finite set of Z do not give the Mellin integral's convergence at infinity. A multiplier allowed to vanish could cancel a reciprocal pole. These are structural requirements, not details repairable by numerical positivity.

4.2 A bounded truncation guarantee

Lemma 4.2 (Gaussian height truncation)**.** *Suppose $|H_\eta(2 + \mathrm{i}t)/L_F^S(2 + \mathrm{i}t, \eta)| \leq M(1 + |t|)^A$ for $A \geq 0$. Let $f_{\eta,T}$ denote (4.3) truncated at $|t| \leq T$. If $T \geq \max(1, \sqrt{A})$, then*

$$|f_\eta(Z) - f_{\eta,T}(Z)| \leq \frac{2^A M e^{49/36}}{\pi} Z^{2+c} T^{A-1} e^{-T^2}.$$

Proof. Taking absolute values bounds the tail by $Me^{49/36}Z^{2+c}\pi^{-1}\int_T^\infty (1+t)^A e^{-t^2}\,dt$. For $t \ge T \ge 1$, $(1+t)^A \le 2^A t^A$. Since $T^2 \ge A$, integration of $A/u \le u$ on $[T,t]$ gives $t^A \le T^A e^{(t^2-T^2)/2}$. Finally $\int_T^\infty e^{-t^2/2}\,dt \le T^{-1}e^{-T^2/2}$, by comparison with $T^{-1}te^{-t^2/2}$. Combining the inequalities proves the bound. □

This controls truncation only. A numerical quadrature needs its own error bound, and evaluating a signal at finitely many scales does not establish (4.4). The lemma is useful for reproducible diagnostics, not an analytic continuation certificate from finite data.

5 Completed theta reflection: what can be checked locally

For clarity, the arithmetic conventions are $F = \mathbb{Q}(\sqrt{-3})$, $\omega = e^{2\pi \mathrm{i}/3}$, $\mathcal{O} = \mathbb{Z}[\omega]$, and $\lambda = \sqrt{-3} = 1+2\omega$. An ideal coprime to 3 is represented by its unique primary generator congruent to 1 (mod 3). Write $q_a = |a|^2$ and $\alpha(a) = a/|a|$ for $a \neq 0$. The fixed excluded set S contains the primes above 6 and every prime in the defining moduli of the fixed, zero-extended character presentations. For a good prime p, $\chi_p(a)$ is the sixth root of unity congruent to $a^{(q_p-1)/6}$ (mod p) on units and is zero otherwise. Extend it multiplicatively in the denominator. All integer powers, including zero and negative powers, remain zero on nonunits. For squarefree c,

$$e(z) = \exp\bigl(2\pi\mathrm{i}\,\mathrm{Tr}_{F/\mathbb{Q}}(z/\lambda)\bigr), \qquad \gamma_j(c) = q_c^{-1/2} \sum_{v \bmod c} \chi_c(v)^j e(v/c), \quad \gamma_j(1) = 1.$$

These are the conventions actually read in source lines 566–645; in particular, γ_2 is a normalized cubic Gauss sum.

The source's completed Dirichlet product is

$$T(s,\Psi) = \left(\sum_n^{\mathrm{sf}} \gamma_2(n)\bar{\alpha}(n)\Psi(n)q_n^{-s}\right)\left(\sum_b \bar{\alpha}(b)^3\Psi(b)^3 q_b^{-3s+1/2}\right).$$

Here $q_n = |n|^2$, $\alpha(n) = n/|n|$, and the primary and fixed-unit conditions of the source remain in both sums. Absolute convergence for $\Re s > 1$ allows multiplication. At a common prime the valuation in nb^3 is still allowed; forcing coprimality would change this product. The smoothing has coefficient $q_n^{-1/2}q_b^{-1}$ because the Mellin power $X^{s-1/2}$ evaluates at $q_n q_b^3$.

5.1 The exponent-zero mask is not the constant function

Let χ be an order-six character on a finite field of size Q and let e_Q be a nontrivial additive character. Every power is extended by zero, including $\chi^0 = 1_{x\neq 0}$. Its normalized finite Fourier coefficient is

$$C_j(h) = Q^{-1}\sum_x \chi(x)^j e_Q(-hx).$$

For $j \not\equiv 0 \pmod 6$ and $h \neq 0$, substitution $y = hx$ gives $C_j(h) = Q^{-1/2}\tau_j^-\chi(h)^{-j}$, with $\tau_j^- = Q^{-1/2}\sum_y \chi(y)^j e_Q(-y)$. Multiplicative orthogonality gives $C_j(0) = 0$. For $j = 0$, additive orthogonality instead gives

$$C_0(0) = 1 - Q^{-1}, \qquad C_0(h) = -Q^{-1} \quad (h \neq 0). \tag{5.1}$$

Thus erasing an exponent-zero factor loses both a nonzero-frequency branch and the scalar $1 - Q^{-1}$. This directly explains the source's active/inactive prime distinction.

The reflected multiplier includes a further $\chi(h)^{-2}$. With $y = h^{-1}$, the active local sum becomes a character of exponent $j+2$. At $j=4$ it is the Ramanujan sum $\sum_{y\neq 0} e_Q(\epsilon xy) = -1 + Q1_{x=0}$, not a unit-modulus character. For active $j=0$, the factor $-Q^{-1}$ in (5.1) produces a size $Q^{-1/2}$ cubic Gauss term. These finite deductions agree with `eq:reflection-local-fourier` and the displayed B_p branches in the source. They do not verify the whole cusp construction.

5.2 The analytic normalization and its dependency boundary

The source differentiates a finite translated theta sum in $\bar{z}$, then takes a Mellin integral against $v^{2s-1}\,dv$. For a cusp expansion $C_H(v) + \sum_{\mu\neq 0} d_H(\mu) v K_{1/3}(4\pi|\mu|v) e^{2\pi \mathrm{i}(\mu z + \bar{\mu}\bar{z})}$, differentiation kills C_H. The classical Bessel integral

$$\int_0^\infty K_{1/3}(u) u^{w-1}\,du = 2^{w-2}\Gamma((w-1/3)/2)\Gamma((w+1/3)/2), \quad \Re w > 1/3,$$

with $w = 2s+1$, yields gamma factors $\Gamma(s+1/3)\Gamma(s+2/3)$. The source's infinity-index substitution $\mu = \lambda^{-3} n b^3$, $q_{\lambda^{-3}} = 1/27$, gives its scalar

$$\frac{3^{5/2}}{4}\left(\frac{27}{(2\pi)^2}\right)^s \Gamma(s+1/3)\Gamma(s+2/3) T(s,\Psi).$$

The reflected change $v \mapsto 1/(q_c v)$ yields $\Gamma(3/2 - s \pm 1/6)$; setting $t = 1/2 - s$ gives

$$R(t) = \prod_{\pm} \frac{\Gamma(1+t\pm 1/6)}{\Gamma(1-t\pm 1/6)}, \qquad K = \frac{(2\pi)^4}{27}.$$

The first negative numerator pole is $-5/6$, so the test-kernel shift to $\Re t = -1/4$ crosses no such pole. This checks the local Mellin bookkeeping and the reason the theta constant term must be removed.

Dunn–Radziwiłł's actual cusp section supplies automorphy and coefficient formulas; its prime Gauss-sum asymptotic is GRH-conditional [3]. The source expressly does not import that conditional asymptotic, and constructs its own finite translated masks rather than invoking a unit-Fourier-support twist formula for masks which may fail that support condition. A complete verification must still prove every rational-cusp multiplier, common-sector convention, dual support bound, and the claimed uniform norm estimates as moving primes vary. The entireness argument also needs decay at *both* ends of the vertical coordinate after differentiating. Our finite mask calculation does not substitute for any of these obligations.

6 Short primes, bootstrap hypotheses, and principal normalization

6.1 Two prime inputs that must not be confused

The fixed-ray normalizer is a consequence of the prime ideal theorem in one fixed abelian extension. For a fixed quotient T, a fixed nonnegative nonzero annular W, and $P \to \infty$, it is

$$\sum_{p\in 1_T} W(q_p/P) q_p^{-5/6} \sim \frac{P^{1/6}}{|T|\log P}\int_0^\infty W(u) u^{-5/6}\,du. \tag{6.1}$$

To see the weighted implication, write the sum as a Stieltjes integral against $\pi_{1_T}(x) \sim \mathrm{Li}(x)/|T|$. On $[aP, bP]$ containing the support, the remainder is uniformly $o(P/\log P)$. Integration by parts costs a weight derivative of order $P^{-11/6}$ over length $O(P)$ and boundary terms of order $P^{-5/6} o(P/\log P)$;

both are $o(P^{1/6}/\log P)$. Substitution $x = Pu$ in the main integral gives (6.1), since $\log P/\log(Pu) \to 1$ uniformly. Fixed excluded primes eventually leave the window. No conductor-uniform threshold follows. Thorner–Zaman provides the relevant fixed-extension framework while its uniform zero-free theory allows an exceptional real zero [6].

The second prime input concerns *moving nonprincipal characters* and normalized products of short prime sums. It is not implied by (6.1). Here the source assumes

$$\beta_* \le s_\kappa = (1+\kappa)/2 < 1, \tag{6.2}$$

and uses a logarithmic-derivative bound on the strictly rightward line $\Re s = s_\kappa + e$ for every fixed $e > 0$. The contour for a smoothed von Mangoldt sum starts in absolute convergence. If the inducing character is nonprincipal and (6.2) and the stated logarithmic control hold, the shift crosses no pole and gives a bound of size $P^{s_\kappa+e}$ times a conductor logarithm and profile/height factors. Higher prime powers cost $O(P^{1/2+\epsilon})$. Dividing the annular profile by $\log(Pu)$ removes the prime logarithm for large P.

After normalization by $P^{-1/2}$ and squaring, the exponent is $2s_\kappa - 1 + 2e = \kappa + 2e$. A fixed list of slots of lengths z_i therefore has squared bound $Z^{\kappa\sum_i z_i+\epsilon}$, with finite smooth seminorm and height factors still present. A polynomial bound on moving conductor norms converts their logarithms into an arbitrarily small power; it is not permissible to suppress arbitrary conductor growth. Deleting $O(\log Z)$ primes from a polynomially bounded radical costs $O(P_i^{-1/2}\log Z)$ per normalized slot and must be included in the requested loss.

The source's finite group Θ contains all coefficient characters ϑ_{ij}. If a row induces $\psi \notin \Theta$, then $\psi\vartheta_{ij}$ cannot be principal. This is why the positive-slot row family excludes Θ, not merely the principal character. A principal slot would have a residue of volume size and invalidate the stated nonprincipal prime argument.

At the actual choice $\kappa = 2\beta_* - 1$, (6.2) is an equality. This is a valid *formal bootstrap premise*: the shifted line is to the right of the family supremum. It does not assume $\beta_* \le 7/8$. However, the upper restriction $\kappa \le 5/6$ still depends on the first-stage theorem, and the required logarithmic bounds, finite induction, and uniformity have to be proved. The apparent equality does not certify those analytic estimates.

6.2 The two moment contracts

At base $Z > 1$, for a smooth annular profile W, the plain polynomial is

$$S_\psi(n;W) = Z^{-n/2}\sum_l \psi(l)W(q_l/Z^n), \qquad Q_{\psi,i} = Z^{-z_i/2}\sum_{p\ \text{prime}} \psi(p)\nu_i(p)W_i(q_p/Z^{z_i}).$$

Sums use good ideals with their primary generators. Every zero mask in ψ is retained. A real log-length denotes the norm scale Z^n or Z^{z_i}, not a bound on a complex embedding. The source's marked inverse polynomial is $M_u(Z^r) = Z^{-r/2}\sum_n \mu(n)\psi_u(n)W(q_n/Z^r)$. For disjoint, row-independent prime supports and a common fixed twist in all factors, its claimed bound is

$$\sum_{0<q_u\ll Z^m} |M_u(Z^r)Q_u|^2 \ll Z^{m+\epsilon}, \quad r + 2z \le m - c_1, \quad 2r + 8z \le 3m - c_2.$$

The constants c_1, c_2 must remain positive before the moving scale. At vanishing inverse capacity the source uses a distinct unmarked amplification estimate; one cannot take a limiting strict-margin theorem with uncontrolled constants.

For two plain polynomials the claimed mean square is $\sum_{k\in\mathcal{R}_z}|S_{\psi_k}(n_1)S_{\psi_k}(n_2)Q|^2 \ll Z^{M+\epsilon}$, with unrestricted bounded lengths when $z=0$, but $n_1+n_2+6\kappa z\le M$ and (6.2) when $z>0$. Both factors and all slots have the same puncture mask. The mesh must depend only on prescribed real ranges and loss, uniformly in $\kappa\in[3/4,1]$, before a fixed slot count is installed. Its seminorm/height orders may then depend on that count and fixed arithmetic data. The source's centered differences use equal products of scales and the same profiles to cancel certain main terms; that is not a claim of cancellation in every uncentered exceptional row.

6.3 Normalizer and residue bookkeeping

For fixed positive ℓ_i summing to ℓ, write $P_i=Z^{\ell_i}$ and let $S_i(Z)$ denote the left side of (6.1) in slot i. Then the source chooses

$$A_T(Z)=(-1)^K Z^{-\ell/6}\prod_{i=1}^{K}S_i(Z)\asymp(\log Z)^{-K}$$

in absolute value, with nonzero leading constant. Even K makes its eventual sign positive. The fixed tuple factorization at the principal residue gives a product of factors $-1+O(q_p^{-7/8})$, and hence a relative error $O(Z^{-\kappa_P})$ for any $0<\kappa_P<(7/8)\min_i\ell_i$. This requires fixed K, nonnegative weights, and uniform local errors, not just an average of signed weights.

The same $c_S A_T(Z)$ divides the physical low sum and the high comparison. The scalar residue at $z=1/6$ of $\zeta_F^S(6z)$ is $1/6$ of the residue of ζ_F^S at one. Omitting that Jacobian changes the signal. The principal contour margins are $m_w=23/960$ and $m_z=13/9600$. The principal approximation error is bounded on its original line; it cannot be treated as a holomorphic correction on new contours without proof. These facts expose the exact normalization obligations behind Proposition 4.1.

7 Rederiving the finite endpoint problem

Let $\delta=2a-1$, $q\in[0,\delta/2]$, $x=q/\delta$ when $\delta>0$, and $y=1/2-x$. The closed algebraic domain is

$$\mathcal{B}=[0,5/6]\times[0,1/2]\quad\text{in }(\delta,y).\tag{7.1}$$

At $\delta=0$ the rational formulas below are their continuous algebraic extension; no actual detector ratio q/δ is asserted there. The operative source range has $\delta>1/50$, with a separate floor.

For a row-count exponent R and frequency exponent d, the source's `eq:common-high-exponent` is

$$\begin{aligned}E(d)&=a-\frac{7}{8}+h\left(\frac{17}{50}-\frac{1}{6}\right)-al_y-(1-a)\ell-(\delta/2-q)\ell+d\left(R+\delta/2-\frac{17}{50}\right)\\&=-\frac{1}{48}+\frac{2}{3}\delta+\frac{q}{6}-h(1-R)+(d-h)\left(R+\delta/2-\frac{17}{50}\right).\end{aligned}\tag{7.2}$$

Substitute $a=(1+\delta)/2$ and (2.3) to check equality. In particular $q/6$ is a base-Z exponent, not $dq/6$.

Put $\alpha=5/6$ and

$$D=3-\frac{17x}{9}=\frac{37+34y}{18},\quad P=\left(2-\frac{8x}{9}\right)(1-x)=\frac{7+18y+8y^2}{9}.\tag{7.3}$$

The two ideal row envelopes are

$$R_s(t) = 1 - \delta + \frac{\delta P}{D}(3/2 - t), \qquad L(t) = 1 - \delta + (\alpha - \delta)(t - 1). \tag{7.4}$$

The former is decreasing and the latter increasing for $0 < \delta < \alpha$. Therefore their maximum on $[1, 3/2]$ is minimized at their unique crossing. Solving gives

$$\mathcal{J} = (\alpha - \delta)D + \delta P, \quad t_* = 1 + \frac{\delta P}{2\mathcal{J}}, \quad R_* = 1 - \delta + \frac{(\alpha - \delta)\delta P}{2\mathcal{J}}. \tag{7.5}$$

One has $1 < t_* < 3/2$. At $\delta = 0$, $t_* = 1$ is a minimizer; at $\delta = \alpha$, $t_* = 3/2$ is a minimizer and $R_* = 1 - \delta$. Thus the source's adaptive cutoff is the exact minimax cutoff for *these two affine envelopes*, not necessarily for all possible analytic row estimates.

Write

$$j(\delta, y) = 185 + 170y + (-138 + 12y + 96y^2)\delta, \tag{7.6}$$

$$N(\delta, y) = 370 + 340y - 8(237 + 377y + 26y^2)\delta + 48(51 + 131y + 94y^2 + 32y^3)\delta^2. \tag{7.7}$$

Expanding (7.3) gives $\mathcal{J} = j/108$. On (7.1), $\partial_y j = 170 + (12 + 192y)\delta \geq 170$ and $\partial_\delta j = -138 + 12y + 96y^2 \leq -108$. Consequently

$$70 = j(5/6, 0) \leq j(\delta, y) \leq j(0, 1/2) = 270. \tag{7.8}$$

In particular $35/54 \leq \mathcal{J} \leq 5/2$, so every denominator clearing below preserves inequalities.

Set $E_* = E(h)$ with $q = (1/2 - y)\delta$ and $R = R_*$. Direct substitution first gives

$$G(\delta, y) := -E_* = \frac{1 + (3 + 8y)\delta}{48} - \frac{13}{32}\frac{(5/6 - \delta)\delta P}{\mathcal{J}}.$$

Multiplication by $96j$ and expansion yields the exact rational form

$$G(\delta, y) = \frac{N(\delta, y)}{96 j(\delta, y)}. \tag{7.9}$$

The sign convention is important: we minimize the *positive margin* G, not the negative exponent E_*. The parent code's variable named x denotes our δ, not the source's amplitude ratio x.

7.1 Checking the source's original certificate

Let $v = 51 + 41y$. Polynomial expansion gives

$$\begin{aligned} vN = {} & (3 + 5y)\{(4v\delta - 79)^2 + 49\} \\ & + 4y\{4v\delta[(1 + 3y)(15 + 32y)\delta + 9 - 13y] + 265 + 3485y\}. \end{aligned} \tag{7.10}$$

All factors on the second line are nonnegative on (7.1), because $9 - 13y \geq 5/2$. Since $v \leq 17(3 + 5y)$,

$$N \geq 49/17, \qquad G \geq \frac{49}{17 \cdot 96 \cdot 270} = \frac{49}{440640}.$$

This verifies the source's stated finite inequality. The proof is deliberately conservative: it discards the square and replaces j by a global upper bound, which need not be simultaneous equalities.

8 The exact minimum and all 39 Bernstein coefficients

8.1 The tensor certificate rule

For $0 \le k \le n$,

$$u^k = \sum_{i=k}^{n} \frac{\binom{i}{k}}{\binom{n}{k}} B_i^n(u), \qquad B_i^n(u) = \binom{n}{i} u^i (1-u)^{n-i}.$$

Indeed the right side is $u^k(u+1-u)^{n-k}$ after the binomial coefficient cancellation. Apply the identity in each variable. If $p(u,v) = \sum_{k,l} a_{kl} u^k v^l$ has bidegree at most (n,m), its tensor Bernstein coefficient is

$$b_{ij} = \sum_{k \le i,\ l \le j} a_{kl} \frac{\binom{i}{k}}{\binom{n}{k}} \frac{\binom{j}{l}}{\binom{m}{l}}. \tag{8.1}$$

The tensor weights $B_i^n(u) B_j^m(v)$ are nonnegative and sum to one on $[0,1]^2$. Thus $\min b_{ij} \le p \le \max b_{ij}$. On a rectangle first substitute $\delta = a + (b-a)u$, $y = c + (d-c)v$. All arithmetic below is rational. Closed faces are included, and common boundaries between rectangles require no limiting numerical convention.

8.2 Excluding the low-delta region

Let $P_0 = N - (96/2000)j$. On the rectangles $B_1 = [0, 1/6] \times [0, 1/2]$ and $B_2 = [1/6, 1/3] \times [0, 1/2]$ its bidegree-$(2,3)$ Bernstein arrays are respectively

$$\begin{pmatrix} 9028/25 & 31232/75 & 7076/15 & 13176/25 \\ 25459/125 & 244217/1125 & 257642/1125 & 29934/125 \\ 14278/125 & 14356/125 & 138334/1125 & 17988/125 \end{pmatrix} \tag{8.2}$$

and

$$\begin{pmatrix} 14278/125 & 14356/125 & 138334/1125 & 17988/125 \\ 3097/125 & 14191/1125 & 2114/125 & 6042/125 \\ 416/125 & 8428/1125 & 53468/1125 & 18096/125 \end{pmatrix}. \tag{8.3}$$

Rows index the δ degree and columns the y degree after affine normalization. Every entry is positive. Their respective minima are $14278/125$ and $416/125$. Hence

$$G(\delta, y) > 1/2000 \quad (0 \le \delta \le 1/3,\ 0 \le y \le 1/2). \tag{8.4}$$

This is stronger than a nonnegative test at selected grid points: the grid in the independent verifier reconstructs the unique polynomial coefficients, and the Bernstein convex-hull theorem supplies the continuum conclusion.

8.3 Monotonicity in the upper rectangle

Polynomial differentiation gives

$$N_y j - N j_y = 8\delta Q(\delta, y), \qquad G_y = \frac{\delta Q}{12 j^2}, \tag{8.5}$$

where

$$\begin{aligned} Q = {} & \delta^2(18432y^4 + 4608y^3 - 148176y^2 - 214416y - 112140) \\ & + \delta(65280y^3 + 238320y^2 + 261360y + 148260) \\ & - 8500y^2 - 18500y - 35875. \end{aligned}$$

On $B_3 = [1/3, 5/6] \times [0, 1/2]$, its bidegree-$(2,4)$ Bernstein array is

$$\begin{pmatrix} 1085 & 13369/2 & 87323/6 & 25389 & 40014 \\ 19460 & 28760 & 125350/3 & 59760 & 84240 \\ 9800 & 16100 & 78100/3 & 41400 & 64800 \end{pmatrix}. \tag{8.6}$$

Its minimum is $1085 > 0$, so $G_y > 0$ throughout B_3. The two low rectangles and B_3 cover (7.1) exactly with disjoint interiors. Equations (8.2), (8.3), and (8.6) give all $12+12+15 = 39$ coefficients.

Theorem 8.1 (Exact finite endpoint minimum)**.** *On $\mathcal{B}$ the unique minimum of $G = N/(96j)$ is*

$$\gamma = \frac{-4473 + 35\sqrt{16354}}{12696}, \quad \textit{attained at} \quad (\delta_*, y_*) = \left(\frac{185}{138} - \frac{35\sqrt{16354}}{4692}, 0 \right). \tag{8.7}$$

Moreover $1/4384 < \gamma < 1/4383 < 1/2000$ and $1/3 < \delta_* < 3/4$.

Proof. By the preceding certificates the low region has margin greater than $1/2000$, and the upper region has its minimum in y uniquely at zero. There

$$G(\delta, 0) = \frac{370 - 1896\delta + 2448\delta^2}{96(185 - 138\delta)}.$$

An explicit one-variable derivative check is

$$\frac{d}{d\delta} G(\delta, 0) = -\frac{3(9384\delta^2 - 25160\delta + 8325)}{8(185 - 138\delta)^2}. \tag{8.8}$$

The roots of its numerator are δ_* and $\delta^+ = 185/138 + 35\sqrt{16354}/4692 > 5/6$. The quadratic has positive leading coefficient, so the derivative is negative before δ_* and positive between δ_* and $5/6$. The endpoint values are

$$G(0,0) = \frac{1}{48}, \qquad G(1/3, 0) = \frac{5}{6672}, \qquad G(5/6, 0) = \frac{7}{96},$$

all strictly greater than $1/4383$. Thus no boundary value competes with the claimed minimum. The remaining boundary with $y > 0$ in the upper rectangle is larger by strict y-monotonicity; the low-region bound covers its other edges. Direct coefficient comparison in $\mathbb{Q}(\sqrt{16354})$ gives

$$N(\delta, 0) - 96\gamma j(\delta, 0) = 2448(\delta - \delta_*)^2. \tag{8.9}$$

For clarity, the quadratic coefficient is 2448; the linear coefficient condition is $-1896 + 13248\gamma = -4896\delta_*$; and the constant condition is $370 - 17760\gamma = 2448\delta_*^2$. These three identities verify the square without a root-finder.

The number γ is the positive root of $304704t^2 + 214704t - 49 = 0$. The polynomial is strictly increasing for $t \geq 0$ and has opposite signs at $1/4384$ and $1/4383$. Thus the claimed bracket holds. Positivity can also be seen from $1225 \cdot 16354 - 4473^2 = 1029 > 0$. The square-root bounds $127 < \sqrt{16354} < 128$ imply $1/3 < \delta_* < 3/4$ directly from (8.7). Since $j > 0$, (8.9) gives the unique minimum on $y = 0$. It lies in the upper rectangle, and its value is below the strict low-region lower bound. Strict monotonicity in y proves global uniqueness on (7.1). □

For orientation, $\gamma \approx 0.00022814730881969814$ and $\delta_* \approx 0.38663756218397807$. These decimals are not evidence for the proof. The usable rational bound $1/4384$ improves the source constant by the exact factor

$$\frac{1/4384}{49/440640} = \frac{13770}{6713} > 2.$$

The minimizer lies above $1/50$ and below $3/4 \le \kappa$ for every admissible dynamic κ. Therefore the exact minimum is not an artifact of extending the algebraic box beyond the live upper bin ceiling. This fact still says nothing about whether the analytic row count actually realizes the hypothesized exponent envelope.

9 New bounded stability and perturbation results

Proposition 9.1 (Quantitative localization of near minimizers)**.** *For $1/3 \le \delta \le 5/6$ and $0 \le y \le 1/2$,*

$$G(\delta, y) - \gamma \ge \frac{51}{278}(\delta - \delta_*)^2 + \frac{217}{524880}y. \tag{9.1}$$

If $0 \le \varepsilon < 1/2000 - \gamma$ and $G(\delta, y) \le \gamma + \varepsilon$ on $\mathcal{B}$, then necessarily

$$\delta > 1/3, \qquad |\delta - \delta_*| \le \sqrt{278\varepsilon/51}, \qquad y \le (524880/217)\varepsilon.$$

Proof. On B_3, $Q \ge 1085$, $\delta \ge 1/3$, and $j \le 270$, so (8.5) gives $G_y \ge 1085/(36 \cdot 270^2) = 217/524880$. Integrate from 0 to y. On $y = 0$ the square identity gives

$$G(\delta, 0) - \gamma = \frac{51}{2(185 - 138\delta)}(\delta - \delta_*)^2 \ge \frac{51}{278}(\delta - \delta_*)^2,$$

because $185 - 138\delta \le 139$ when $\delta \ge 1/3$. Add the bounds. The strict low-region exclusion (8.4) then proves the near-minimizer assertions. □

This is a useful reduction for later geometry searches: a small endpoint margin can occur only near the single boundary point identified above. The constants are explicit, but intentionally not claimed optimal. They bound the fixed source rational function, not a perturbed arithmetic estimate whose error has not been controlled.

Proposition 9.2 (Certified perturbation tolerance)**.** *Let $\widetilde{N} = N + e_N$ and $\widetilde{j} = j + e_j$ be real functions on $\mathcal{B}$ with $|e_N| \le \varepsilon_N$ and $|e_j| \le \varepsilon_j < 70$. For $0 < \tau < \gamma$, if*

$$\varepsilon_N + 96\tau\varepsilon_j < 6720(\gamma - \tau), \tag{9.2}$$

then $\widetilde{N}/(96\widetilde{j}) > \tau$ throughout $\mathcal{B}$. In particular the entirely rational conditions $\varepsilon_N \le 1/10$, $\varepsilon_j \le 1$ imply $\widetilde{N}/(96\widetilde{j}) > 1/5000$.

Proof. We have $\widetilde{j} \ge 70 - \varepsilon_j > 0$. By Theorem 8.1,

$$\widetilde{N} - 96\tau\widetilde{j} \ge 96j(\gamma - \tau) - \varepsilon_N - 96\tau\varepsilon_j \ge 6720(\gamma - \tau) - \varepsilon_N - 96\tau\varepsilon_j > 0.$$

For the numerical example it suffices to replace γ on the right by $1/4384$. The exact rational inequality $1/10 + 96/5000 < 6720(1/4384 - 1/5000)$ is checked directly. □

This result turns independently bounded algebraic approximation errors into a continuum certificate. It does not license unspecified floating point error or a modelling change. In particular perturbing an exponent formula is different from proving that a number-theoretic row sum obeys that formula.

10 A sharper dynamic capacity penalty

The source's plain capacity is $z_P(m) = (1-2m)/(6\kappa)$, whereas its baseline calculation uses $2(1-2m)/9$, the value at $\kappa = 3/4$. The resulting increase in the row exponent is

$$\mathcal{P}(\Delta, q, m) = 2q(1-2m)\left(\frac{2}{9} - \frac{1}{9/2+12\Delta}\right) = \frac{24q(1-2m)\Delta}{(9/2)(9/2+12\Delta)}. \tag{10.1}$$

Its domain comes from the *actual* short-witness branch of the source, not from arbitrary nonnegative q, m.

Proposition 10.1 (Dynamic capacity improvement)**.** *Suppose* $0 \le \Delta \le 1/24$, $\kappa = 3/4 + 2\Delta$, $0 \le \delta \le \kappa$, $0 \le q \le \delta/2$, *and* $1/3 \le m \le 1/2$. *Then*

$$0 \le \mathcal{P}(\Delta, q, m) \le \frac{4}{27}\Delta. \tag{10.2}$$

For $m \ge 1/3 - \varepsilon_m$ *in the positive-capacity branch, the upper bound becomes* $4\Delta/27 + (8/9)\Delta\varepsilon_m$. *The coefficient* $4/27$ *is sharp for these relaxed scalar constraints when* $\Delta > 0$.

Proof. All factors in (10.1) are nonnegative on the stated range. The expression is increasing in q and decreasing in m. Retain $q \le \kappa/2$, instead of replacing it by the weaker $q \le 1/2$. Since $9/2 + 12\Delta = 6\kappa$,

$$\mathcal{P} \le \frac{24(\kappa/2)(1/3)\Delta}{(9/2)6\kappa} = \frac{4}{27}\Delta.$$

Replacing $1/3$ by $1/3 + 2\varepsilon_m$ gives the second assertion. Equality holds at $\delta = \kappa$, $q = \kappa/2$, $m = 1/3$, which proves sharpness on this scalar domain. These parameters are not asserted to be simultaneously realized by an actual arithmetic bin or by the optimal adaptive cutoff. □

In the source's comparison, the short plain branch has $m \ge t - r_*(t) - O(\epsilon) \ge 1/3 - O(\epsilon)$. The exact source bound $\delta \le \kappa$ is essential to the improved coefficient. If one discards that relation and keeps only $\delta \le 5/6$, the same argument no longer proves $4/27$. Other witness, selection, and mesh errors retain their own costs; Proposition 10.1 sharpens only this capacity replacement.

Corollary 10.2 (Conditional endpoint bridge)**.** *Assume the source's moment and detector hypotheses actually establish the row envelope with the capacity replacement in Proposition 10.1, so that* $R \le R_* + 4\Delta/27 + \varepsilon_{\text{row}}$. *Assume also its high-exponent interface (7.2), with all other real and retained-height costs included in* $\varepsilon_{\text{total}}$. *Then at* $d = h$,

$$E_{\text{actual}}(h) - \Delta \le -\gamma - \frac{95}{108}\Delta + \varepsilon_{\text{total}}. \tag{10.3}$$

Keeping the source's original capacity bound instead gives $-\gamma - (51/64)\Delta + \varepsilon_{\text{total}}$.

Proof. The coefficient of R in (7.2) at $d = h$ is $h = 13/16$. Theorem 8.1 gives $E_* \le -\gamma$. Therefore $E_{\text{actual}}(h) - \Delta \le -\gamma - (1 - 4h/27)\Delta + \varepsilon_{\text{total}}$, and $1 - 4h/27 = 95/108$. For the original bound, $1 - h/4 = 51/64$. □

The subtraction of Δ is indispensable: the high comparison is measured relative to $C(\beta_*)$, not $C(7/8)$. Neither corollary proves $E_{\text{actual}}(h) < 0$ for every Δ. The savings are local to the adaptive endpoint and cannot simply be substituted for the source's global minimum of all margins.

11 Other ranges and uniform loss accounting

The endpoint is only one of several frequency ranges. The source also has the following error-free bounds with the fixed geometry.

Range	Ideal margin or exponent	Qualification
Floor $\delta = 1/50$	$E(h) \le -7/1200$	No actual zero witness is required for every floor row.
$d \le 1/2$, moderate	$E(d) \le -49/14400$	Uses the unselected envelope $R = 76/75 - 2\delta/3$.
Closed $\delta = 5/6$	$E(h) \le -1/48 - \delta/16$	Uses unmarked amplification across $r = 1$, not shrinking strict marked margins.
Small $d \le 1/100$	Saving at least $63/800$ relative to $C(\beta_*)$	Uses full absolute correction and no buffered detector.
Principal residues	$m_w = 23/960$, $m_z = 13/9600$, $\kappa_P > 0$	Normalizer and local principal approximation must hold.
Large $d > h + \zeta$	Arbitrary fixed power saving after choosing z_∞	Uses absolute lines and $\zeta > 0$ fixed in advance.

For example the floor bound follows from $R = 1$, $q \le \delta/2$ in (7.2). At $d = 1/2$, substitution of the indicated R gives $E(1/2) \le -529/2400 + (25/96)\delta$, which is at most $-49/14400$ for $\delta \le 5/6$. For small rows, $h(17/50 - 1/6) - l_y/2 = -79/800$; adding the conservative $2d_{\min} = 1/50$ gives $-63/800$.

For $R \ge 1 - \delta$, $R + \delta/2 - 17/50 \ge 33/50 - \delta/2 > 0$ on the relevant domain. Thus $d = h$ controls $d \le h$ for those chosen envelopes. The source extends to $h + \zeta$ with slope bounded by two, at cost 2ζ, while the prime supply remains above $1/5 > 7/37$ if $\zeta < 1/48$. The exact margin at $d = h$ does not remove that supply constraint. A change to h, ℓ, or the proposed zero-free boundary requires redoing the principal, low, small-row, and supply calculations, not just the scalar minimum.

Lemma 11.1 (Late height closure)**.** *Let $m > 0$ and b be independent of a target, and suppose for that target there are finite $A_\eta \ge 0$, $B_\eta \in \mathbb{R}$, and a ceiling $\tau_{0,\eta} > 0$ such that for every fixed admissible $\tau > 0$, every fixed integer N, and all sufficiently large Z,*

$$|J_\eta - f_\eta| \ll_{\eta,N,\tau} Z^{b-m}(1 + Z^\tau)^{A_\eta} + Z^{B_\eta - \tau N}.$$

Assume A_η, B_η are fixed before N, and that J_η, f_η themselves do not depend on τ or N. Then $|J_\eta - f_\eta| \ll_\eta Z^{b-m/2}$ after valid choices of these parameters and a target-dependent threshold. There is no loss in the nonnegativity convention: an initially negative height exponent can be replaced by $\max(0, A_\eta)$ in the upper bound.

Proof. Choose $0 < \tau \le \min(\tau_{0,\eta}, m/[4(A_\eta + 1)])$ and then an integer $N > (B_\eta - b + m)/\tau$. The first exponent is at most $b - 3m/4$ and the second is less than $b - m$. Absorb the fixed factor 2^{A_η} and constants into a sufficiently large threshold, leaving the common saving $m/2$. The functions compared did not change during this choice. □

This lemma explains why target-dependent height exponents are compatible with a target-independent real saving. It also exposes a failure mode: if increasing N also increases A_η or the scale degree B_η, this order of choices is not justified. The source's separate internal versus external derivative orders are intended to prevent that dependence; their complete propagation remains a substantive full-verification obligation.

One cumulative height budget is also required. If all buffered arguments have imaginary parts $\gamma_j + \sum_k a_{jk} v_k$, with a fixed finite matrix, choosing $|v_k| \le T_1/[2m_0(1+\max_j |a_{jk}|)]$ for m_0 coordinates makes the total added height at most $T_1/2$. Reusing a fresh $T_1/2$ at every successive transform would not prove this simultaneous condition.

12 Current public primary-literature check

The check was made on 7 October 2026 local time (8 October UTC for some requests). Only public primary sources were requested. No private passage or private report was uploaded or placed in a search query. Full-text theorem passages were inspected where a theorem interface is discussed; recent abstracts are identified as such when used only for positioning.

Density is not exclusion. Guth–Maynard's current arXiv version is v2, revised 7 April 2026; the publisher lists *Annals of Mathematics* 203 (2026), 623–675 [7]. Its zero-density exponent is $N(\sigma, T) \le T^{30(1-\sigma)/13+o(1)}$. For fixed $\sigma < 1$ this upper bound does not tend below one as $T \to \infty$, so it does not exclude every zero to the right of σ. It cannot be substituted for the common-signal contracts.

Cubic theta and sextic sieve inputs. Dunn–Radziwiłł's v3 and published 2024 article distinguish their unconditional cusp formulas from the GRH-dependent prime asymptotic [3]. Blomer–Goldmakher–Louvel's Theorem 1.3 gives the squarefree ideal large sieve with size $(MN)^{\epsilon}(M + N + (MN)^{2/3})$ in its specified higher-order Hecke family [4]. Gao–Zhao's Lemma 2.9 states the sextic-symbol version over Eisenstein integers, with squarefree variables and exclusions at the bad primes [5]. Neither statement alone proves the source's completed-index marked inverse moment or its positive-slot plain recursion; those require the extra mask, multiplicity, and width arguments recorded above.

Nonvanishing proportions. David–de Faveri–Dunn–Stucky's v2, revised 3 March 2026, proves positive-proportion central nonvanishing in a specified primitive cubic Hecke family [8]. This is not all-character nonvanishing in a half-plane. The current abstract of Alpöge–Furman v2 reports an unconditional two-thirds simple critical-line proportion and related improvements [10]; Pearce-Crump's September 2026 preprint reports an optimized positive proportion within Selberg's method [11]. These are relevant comparisons of finite positivity and moment methods, not substitutes for uniform exclusion. Their formalization or broader analytic claims are not independently verified in this report.

Conditional bridges remain conditional. Dong–Wattanawanichkul–Zaharescu's May 2026 preprint studies consequences of appropriately bounded mollified moments of arbitrarily long polynomial length and a family quasi-RH criterion [9]. Its abstract supplies current context for the distinction between a bridge theorem and its moment hypotheses. We do not invoke that criterion here.

Source version and status. The public repository also lists an October 5 alternate 11/12 argument and an exceptional-zero companion. They are different manuscripts, not silent replacements for the pinned September 30 source. The selected source's own README says preprint. The present search found no evidence that would justify promoting this report's bounded verification into acceptance of the 7/8 claim. This is a statement about the scope of this check, not a proof that no subsequent assessment exists.

13 Benefits, limitations, and future work

13.1 What the Ram viewpoint contributes

The logarithmic atom formulation makes the Mellin measure and sign conventions explicit and separates representation from reconstruction. The finite polynomial formulation makes denominator signs, closed endpoints, and error budgets checkable. The Bernstein arrays replace a sampling heuristic by a proof over a continuum rectangle. The exact square identity proves optimality, while the derivative certificate explains why the worst case is on $y = 0$. The dynamic capacity computation retains an actual source constraint and improves a loss coefficient without changing the source geometry. These are concrete benefits.

The price is limited scope. Tensor coefficient counts grow with degree and dimension; subdivision can be expensive near zeros. A negative Bernstein coefficient is inconclusive. Coordinate conditioning matters for measured moments, even when rational symbolic arithmetic is exact. None of these methods creates the needed uniform character moments, controls all arithmetic conductors by itself, or turns a local inequality into a universal zero-exclusion theorem.

13.2 A prioritized next programme

First, extend the completed bounded analytic review by independently rederiving the estimates at its remaining high-burden interfaces: whole-index theta masks and common sectors; sextic family admissibility; simultaneous detector witnesses; both finite-width inductions including exceptional inducing characters; and the full compensated correction before contour subdivision. Each successful verification should name the fixed and moving parameters and preserve all zero extensions.

Second, formalize the bounded certificate in a small proof assistant development over rationals and $\mathbb{Q}(\sqrt{16354})$. The 39 arrays and (8.9) are a compact input. Such a formalization would certify this theorem only; its statement should not mention Hecke zeros without explicit hypotheses connecting them to the finite polynomial.

Third, test a coupled optimization of h, ℓ, l_x, l_y using exact feasibility bounds for every range. The unique-minimizer localization offers a starting region, but feasibility must include low-side energy, prime supply, principal contours, and the minimum slot length. Maximizing only γ is not an optimization of the full analytic saving.

Fourth, extend the perturbation theorem to interval-certified coefficient families produced by such a geometry search. Every proposed error must be accounted for in the numerator and denominator bounds, with strict denominator positivity. Approximate symbolic expressions should be replaced by exact rational enclosures before any theorem is stated.

Finally, maintain the density-versus-exclusion distinction. A sequence of improving finite tests requires a genuine exhaustion theorem and uniform bounds to imply an infinite conclusion. The private TR220 warning and Proposition 4.1 identify different versions of this same quantifier requirement.

13.3 Contribution assessment

The assessment is C2/5. The report proves a source-specific sharp finite minimum with a complete certificate, gives useful quantitative bounded extensions, and improves the scalar dynamic-capacity comparison. The minimum originated in a prior local audit and is not claimed as a fresh discovery by this author pass. Mellin inversion, common-signal analytic logic, and basis conversion are established methods, not major novelty by restatement. The unresolved arithmetic bridge prevents a higher assessment based on a new global theorem. L4 applicability remains a separate classification.

A Provenance and reproducibility

The machine-readable author evidence is kept separately from shared mutable queues. All manual source edits used `apply_patch`. The following local inputs were actually read:

1. The pinned source introduction and continuation criterion; the entire completed-reflection statement and proof; fixed-ray normalizer; marked and plain moment statements; short-prime/induction-order passage; row-count proof; complete final endpoint, frequency, and order-of-choices sections. Source labels in Sections 2–11 identify the particular claims used.

2. Both complete parent scripts `check_quasi_rh_endpoint.py` and `check_quasi_rh_optimal.py`, their full JSON certificates, and `M005-independent-audit.txt`. The scripts were replayed by removing only their top-level output write from the parsed AST; their mathematical assertions ran unchanged and their reconstructed JSON results equalled the existing files.

3. The four private passages identified in Section 3. Their full local paths, line ranges, and SHA-256 hashes are in `_audit/M005-author-provenance.json`. No claim is based merely on their filenames.

The independent verifier is `_audit/M005-author-verify.py`; its output is `_audit/M005-author-exact-tests.json`. It uses the bundled Python and an already installed SymPy package, with bytecode writes disabled. No package or LaTeX installation was performed. It checks:

1. Both source exponent expressions, the cutoff crossing, and $\mathcal{J} = j/108$ by exact symbolic identities.

2. The original completion-of-square certificate and the full derivative identity, not merely selected numerical evaluations.

3. All 39 Bernstein coefficients by a different method from the parent: evaluate on a rational tensor grid and solve the Bernstein evaluation systems by exact Fraction Gaussian elimination.

4. The optimal square in $\mathbb{Q}(\sqrt{16354})$, root polynomial, strict rational brackets, localization constants, perturbation example, and dynamic capacity and decrement identities.

5. Unchanged hashes of all parent inputs, plus the read-only public pin byte comparison.

Exact grid interpolation proves a polynomial identity only after its bidegree is bounded. It is not finite sampling used as a positivity test. The tensor systems are invertible because a degree-n polynomial vanishing at $n+1$ distinct points is zero, applied one variable at a time.

A.1 Completed mathematical reviews

The parent read the full assembled report and accepted the local minimum, localization, perturbation theorem, 4/27 capacity bound, and conditional 95/108 comparison at C2. Its precision correction, $A_\eta \geq 0$ in Lemma 11.1, is incorporated. The parent also independently verified the radical square, derivative, minimizer interval and boundary values, then replayed all author mathematical assertions, including the 39 independently reconstructed coefficients, without changing the inputs. This review concerns the report's bounded and conditional mathematics; it does not endorse the global source theorem.

The separate analytic-number-theory review (Boole) completed a dependency-chain reading from the common-signal criterion through the first 11/12 stage, compensated 7/8 stage, and final Hecke/Dirichlet transfer. It deliberately excluded the endpoint certificate treated here. Its

disposition was: no decisive counterexample, erroneous identity, or invalid inference was established in the portions checked, and no blocking defect was established. This is weaker than independent proof verification. The source supplies arguments for the unresolved estimates; the review does not allege that those arguments are absent or false.

The review independently checked the logical Mellin implication under its contracts, the fixed-relative-width and fixed-ray prime quantifiers, selected finite identities, and the coherent late-height choice provided the stated uniformity holds. It cross-checked the imported cubic, sextic, and quadratic sieve restrictions. Fixed coefficients within each row sum, full zero masks, and fixed slot counts cannot be dropped. Its exact local tests passed 56160 split residue-field cases at norms $7, 13, 19$, 2895 prime-square congruence cases, eight first-Poisson phase cases, and three affine recurrence identities. These are local falsification checks, not estimates for infinite families or a verification of all inert-prime and Eisenstein cusp normalizations.

The remaining burden includes every scalar and interchange in the all-cusp, all-frequency completed reflection; the full detector and local Euler corrections; every inverse- and fourth-moment recursion inequality, fiber, exceptional case and common-profile bound; and exhaustive propagation of internal height orders independently of the later external tail order. The original Heath-Brown cubic-sieve article was not retrieved directly by that review; its explicit statement was checked through Dunn–Radziwiłł's primary-source restatement and related literature. None of these limits is evidence that the corresponding assertion is false.

The actual review, detailed dependency ledger, literature checks, script, and test output were read in full. Their hashes and precise scope are in `_audit/M005-author-review.json`. The completed review therefore supports a *conditional-on-analytic-inputs* closeout, not certification of the full 7/8 theorem. This distinction also governs Corollary 10.2 and the contribution assessment.

A.2 Physical build and frozen evidence

The native editor request returned a queued opening. The native compiler returned *compile-failed*, with the diagnostic *Unable to find standard directories for platform*; this was an environment failure, not a successful native compilation. The existing `_tools/compile_report.py` produced the physical PDF using only this report's own build directory and the pre-existing TeX installation. The first successful export had 21 pages, no undefined references, and no overfull boxes; every page render was visually inspected. Detailed build outcomes are in `_audit/M005-author-build.json`. The frozen manifest records final source, PDF, and author evidence hashes, excluding shared queues, progress files, catalogues, and mutable status. A manifest does not hash itself.

B Compact non-implication checklist

For future readers, the following implications are *not* proved:

$$\begin{gathered} G > 0 \implies \text{marked moment theorem}, \\ \text{many verified finite blocks} \implies \text{all-height or all-conductor control}, \\ \text{positive-proportion nonvanishing} \implies \text{zero-free half-plane}, \\ \text{Mellin representation} \implies \text{the power-saving contracts}, \\ \text{larger endpoint margin} \implies \text{a smaller zero-free boundary}. \end{gathered}$$

The valid route is a conjunction: independently verified arithmetic identities and uniform estimates, correct principal signal and nonvanishing correction, all-range loss accounting, and then the contin-

uation proposition. This report improves and verifies bounded parts of that route. Its conclusions do not constitute a global endorsement of the source's arithmetic or zero-free claims.

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